

Notes: He, Sandomirskiy & Tamuz (JPE, 2026)

Private Private Information

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1 Big picture

- **Question.** How much information about a state can be communicated through signals that are informative about the state but independent of each other?
- **New concept.** The paper introduces *private private* information: several signals each contain information about the state, yet the signals themselves are statistically independent.
- **Motivating privacy problem.** A recommender knows a payoff-relevant state, such as productivity or creditworthiness, and also a sensitive attribute. The recommender wants to be informative about the state while revealing nothing about the sensitive attribute.
- **Core trade-off.** If the state and sensitive attribute are correlated, then a fully revealing recommendation about the state also reveals information about the sensitive attribute. Privacy forces a sacrifice of signal quality.
- **Main optimality notion.** A *private private* structure is Blackwell–Pareto optimal if no signal can be made more Blackwell informative without violating the independence requirement or worsening another signal.
- **Binary-state, two-signal result.** The Blackwell–Pareto frontier has a simple characterization: the belief distributions induced by the two signals must be conjugates.
- **Geometric result.** For binary states and many signals, *private private* structures can be represented by subsets of the unit cube. Blackwell–Pareto optimality corresponds to sets of uniqueness from mathematical tomography.
- **Privacy-preserving recommendation.** With a binary state, there exists a dominant privacy-preserving recommendation: one signal that Blackwell dominates every other privacy-preserving recommendation.
- **Applications.** The results apply to fairness/statistical parity, privacy-constrained information design, causal inference, Bayesian mechanism design, and persuasion in games.
- **Economic takeaway.** Privacy is an information-design constraint, not just a legal or computational constraint. The paper gives a frontier: the most one can learn about the state while learning nothing about another signal.

2 Model objects and measurement

2.1 Information structure and private private signals

Let the state be a random variable

$$\omega \in \Omega = \{0, 1, \dots, m-1\},$$

with full-support prior. There are $n \geq 2$ signals

$$s_1, \dots, s_n.$$

An information structure is

$$I = (\omega, s_1, \dots, s_n).$$

The posterior belief induced by signal s_i is

$$p(s_i)(k) = \Pr[\omega = k \mid s_i], \quad k \in \Omega.$$

For a binary state, $p(s_i)$ is summarized by a scalar in $[0, 1]$:

$$p(s_i) = \Pr[\omega = 1 \mid s_i].$$

Definition. The information structure is *private private* if

$$s_1, \dots, s_n \text{ are independent random variables.}$$

Interpretation: Independence is unconditional. The signals can become correlated after conditioning on the state. This is the opposite of the usual conditionally independent signal model.

2.2 Blackwell informativeness

For a single signal, (ω, s) Blackwell dominates $(\omega, \hat{s})$ if observing s is weakly better than observing $\hat{s}$ in every decision problem. Equivalently, for every convex function φ on the belief simplex,

$$\mathbb{E}[\varphi(p(s))] \geq \mathbb{E}[\varphi(p(\hat{s}))].$$

For two *private private* structures

$$I = (\omega, s_1, \dots, s_n), \quad \hat{I} = (\omega, \hat{s}_1, \dots, \hat{s}_n),$$

write

$$I \succeq \hat{I}$$

if each signal in I Blackwell dominates the corresponding signal in $\hat{I}$.

Interpretation: Blackwell dominance is stronger than a particular utility comparison. It means every possible user of the signal would weakly prefer the more informative signal.

2.3 Blackwell–Pareto optimality

A *private private* structure I is Blackwell–Pareto optimal if, whenever another *private private* structure $\hat{I}$ satisfies

$$\hat{I} \succeq I,$$

then $\hat{I}$ is equivalent to I . Equivalence means that every signal induces the same distribution of posterior beliefs.

Interpretation: This is a robust efficiency frontier. One cannot improve one user’s signal in the Blackwell sense while preserving privacy and not weakening the others.

2.4 Belief distributions

For binary states, each signal s_i induces a random posterior belief in $[0, 1]$. Let

$$F_i(x) = \Pr[p(s_i) \leq x]$$

be the cumulative distribution function of the posterior generated by signal s_i .

Interpretation: The paper studies *private private* structures through the distributions of induced beliefs, not through the labels of the raw signal realizations. This is why direct-revelation versions of signals are sufficient.

3 Models and theoretical specifications

3.1 Conjugate distributions and the two-signal binary frontier

For two binary-state belief distributions F and $\hat{F}$, conjugacy means that the two graphs are reflections around the antidiagonal. In the paper’s notation, the two distributions satisfy a relation of the form

$$F_1(x) = 1 - F_2^{-1}(1 - x),$$

where F_2^{-1} is the generalized inverse.

Theorem 1. For a binary state and two signals, a *private private* structure

$$I = (\omega, s_1, s_2)$$

is Blackwell–Pareto optimal if and only if the distributions of posterior beliefs $p(s_1)$ and $p(s_2)$ are conjugates.

Interpretation: The theorem turns a hard information-design problem into a geometric condition. To check optimality, compare the two posterior-belief distributions. If one signal becomes more informative, the conjugate relation forces the other to become less informative.

3.2 The informativeness trade-off

Corollary 1 states that if two Blackwell–Pareto-optimal structures (ω, s_1, s_2) and (ω, t_1, t_2) satisfy

$$(\omega, t_1) \succeq (\omega, s_1),$$

then

$$(\omega, s_2) \succeq (\omega, t_2).$$

Interpretation: On the Blackwell–Pareto frontier, information is scarce across independent signals. Making the first signal more informative necessarily makes the second less informative.

3.3 Partition representation

The paper shows that every *private private* information structure is equivalent to one generated by a partition of the unit cube. Let the signals be uniform on $[0, 1]^n$ and let

$$A = (A_0, \dots, A_{m-1})$$

be a partition of $[0, 1]^n$. The associated structure sets

$$\omega = k \quad \text{if and only if} \quad (s_1, \dots, s_n) \in A_k.$$

Proposition 1. For every *private private* structure I , there exists a partition A whose associated structure is equivalent to I .

Interpretation: The result lets the paper replace abstract signal spaces with geometry. Signals become coordinates in a unit cube, and states become colored regions of that cube.

3.4 Sets of uniqueness

For binary states, a partition is represented by a single set

$$A \subseteq [0, 1]^n,$$

where $\omega = 1$ if $(s_1, \dots, s_n) \in A$.

A set of uniqueness is a set uniquely determined by its projections onto the coordinate axes. The paper connects this mathematical notion to optimal privacy.

Theorem 2. A binary-state *private private* structure is Blackwell–Pareto optimal if and only if its associated set $A \subseteq [0, 1]^n$ is a set of uniqueness.

Interpretation: The connection to tomography is the paper’s main technical bridge. The projection of A onto coordinate i determines the distribution of beliefs induced by signal s_i . Blackwell–Pareto optimality means that these projections pin down the set itself.

3.5 Lorentz characterization for two signals

Theorem 3, based on Lorentz (1949), characterizes sets of uniqueness in the two-dimensional case. Upward-closed sets in the unit square are key examples. These sets generate Blackwell–Pareto-optimal *private private* structures.

Interpretation: In two dimensions, the geometry is especially tractable. The boundary of an upward-closed set simultaneously describes both posterior distributions, up to reflection. This is why conjugacy emerges.

3.6 Privacy-preserving recommendation

Given an information structure (ω, s_1) , a recommendation s_2 is privacy preserving if it is independent of s_1 . It is optimal if it cannot be Blackwell improved while maintaining independence.

A recommendation s_2^* is dominant if it Blackwell dominates every other privacy-preserving recommendation.

Theorem 4. For any (ω, s_1) with binary state ω , there exists a dominant privacy-preserving recommendation s_2^* . The distribution of $p(s_2^*)$ is the conjugate of the distribution of $p(s_1)$.

Interpretation: This is unusually strong. A designer does not need to know the decision-maker’s objective. The same privacy-preserving recommendation is best for every possible decision problem when the state is binary.

3.7 Constructing the dominant recommendation

For binary ω , the paper gives a simple construction:

1. Calculate $p(s_1) = \Pr[\omega = 1 \mid s_1]$.
2. If $\omega = 1$, draw s_2^* uniformly from $[1 - p(s_1), 1]$.
3. If $\omega = 0$, draw s_2^* uniformly from $[0, 1 - p(s_1)]$.

The constructed signal is independent of s_1 and induces the conjugate belief distribution.

Interpretation: The construction makes the abstract conjugacy theorem operational. It is a recipe for privacy-preserving scoring or recommendation systems satisfying statistical parity.

3.8 Information loss under privacy

The paper discusses information loss with Shannon mutual information. For binary state with prior p , let

$$I(\omega; s) = H(p) - \mathbb{E}[H(p(s))],$$

where H is Shannon entropy. The dominant recommendation satisfies

$$I(\omega; s_2^*) \leq H(p) - I(\omega; s_1).$$

The paper notes that the inequality is strict whenever s_1 is informative but not fully revealing.

Interpretation: Privacy does not merely split information into independent pieces without loss. Some information is destroyed because two independent signals cannot carry the same bit of information about the state.

3.9 Decision value formula

For a decision-maker with indirect utility $U(q)$, Proposition 2 states that the expected payoff from observing the dominant recommendation is

$$\int_0^1 U(1 - F(t)) dt,$$

where F is the CDF of $p(s_1)$.

Interpretation: The formula translates the geometric conjugacy construction into decision value. The privacy loss can therefore be evaluated for any convex indirect utility U .

3.10 Applications beyond recommendation

The paper uses the same structure in several applications:

- **Algorithmic fairness:** independence between recommendation and protected attributes is statistical parity.
- **Information design:** a designer chooses private signals under privacy constraints across receivers.
- **Causal inference:** independent causes generating an effect form a collider, which is a *private private* structure.
- **Mechanism design:** reduced-form feasibility is related to feasible distributions of private posterior beliefs.
- **Zero-sum games:** equilibrium recommendations in games with unique correlated equilibria must be *private private*.

Interpretation: The paper's concept is broad. It is not only a privacy/fairness tool; it also organizes feasibility constraints in several economic models.

4 Identification and theoretical design

4.1 What makes the argument work

- The privacy constraint is exact statistical independence among signals.
- Blackwell dominance gives a decision-problem-free measure of informativeness.
- Posterior belief distributions summarize signals up to Blackwell equivalence.
- Geometry converts signal design into a question about partitions of the unit cube.
- The mathematical literature on sets of uniqueness supplies the frontier characterization.

Interpretation: The paper is not empirical, so credibility comes from the strength and generality of its formal equivalences.

4.2 Why binary states are special

The binary-state case allows posterior beliefs to be represented by a scalar in $[0, 1]$. This makes the conjugacy relation one-dimensional and yields a dominant privacy-preserving recommendation.

Interpretation: For three or more states, posterior beliefs live in a simplex. The Blackwell order is more complicated, and the paper shows that optimal recommendations need not be unique or objective independent.

4.3 What the paper cannot fully settle

- The dominant recommendation result is complete for binary states but not for arbitrary finite state spaces.
- For more than two signals, the set-of-uniqueness characterization is powerful but not always easy to compute.
- The paper characterizes information frontiers but does not estimate privacy costs in a real application.
- Exact independence may be too strong in some applications where approximate privacy is sufficient.
- Implementation of the optimal signals may require knowledge of the joint distribution of the state and protected variables.

Interpretation: These limits do not weaken the main contribution. They clarify where the theory is most directly usable: binary-state recommendation and two-signal privacy constraints.

5 Tables and figures

5.1 Figure 1: Pair of signals uniformly distributed on the unit square

Read:

- Figure 1 shows independent signals (s_1, s_2) uniformly distributed on $[0, 1]^2$.
- The state is binary: $\omega = 1$ in the black region and $\omega = 0$ in the white region.
- The boundary is a diagonal, so observing either signal gives a posterior equal to that signal.
- Both signals are informative, but the signals themselves remain independent.

Economic message: This is the cleanest example of *private private* information: each signal reveals something about the state while revealing nothing about the other signal.

5.2 Figure 2: Binary posterior beliefs with trivial higher-order beliefs

Read:

- Figure 2 modifies the black region so each signal induces posterior beliefs only at $1/4$ or $3/4$.
- The posterior of s_2 has the same distribution even after conditioning on s_1 .
- This means the signals are independent and generate trivial second-order or higher-order beliefs about one another.
- The structure is equivalent to one with binary signals.

Economic message: This example shows that *private private* information can be nontrivial even when each signal takes only two belief values.

PRIVATE PRIVATE INFORMATION

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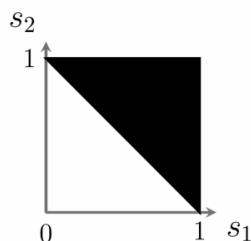


FIG. 1.—Pair of signals (s_1, s_2) uniformly distributed on unit square, with $\omega = 1$ in black area and $\omega = 0$ in white area. Induced posteriors $p(s_1)$, $p(s_2)$ coincide with the signals.

(a) Figure 1: Uniform signals and diagonal state region

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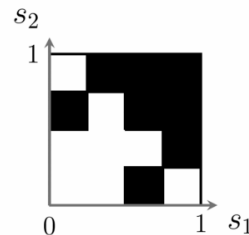


FIG. 2.—Pair of signals (s_1, s_2) uniformly distributed on unit square, with $\omega = 1$ in black area and $\omega = 0$ in white area. Induced posteriors $p(s_1)$, $p(s_2)$ are binary and equally likely to be either $1/4$ or $3/4$. The posterior $p(s_2)$ is equal to $3/4$ on the top half of the square and to $1/4$ on the bottom half. Hence, $p(s_2)$ has the same distribution even conditioned on s_1 , and so the induced second-order and higher-order beliefs are trivial.

DEFINITION 3. We say that a private private information structure $\mathcal{I}$ is Blackwell–Pareto optimal if for every private private information struc-

(b) Figure 2: Binary posterior beliefs

5.3 Figure 3: Example of cumulative distribution function and conjugate distribution

Read:

- Figure 3 plots a belief CDF F and its conjugate $\hat{F}$.
- The two shapes are congruent under reflection around the antidiagonal.
- The CDF F corresponds to a more informative signal because it places less mass near posterior $1/2$.
- The conjugate $\hat{F}$ corresponds to a less informative counterpart.

Economic message: The figure visualizes the central frontier trade-off: a more informative signal for one side forces a conjugate, less informative signal for the other.

5.4 Figure 4: Private private information structure generated by conjugate distributions

Read:

- Figure 4 takes the conjugate distributions from Figure 3 and builds an associated state region in the unit square.
- Signals are uniform on $[0, 1]^2$.
- The state is $\omega = 1$ when $s_2 \geq \hat{F}(1 - s_1)$.
- The induced posterior distributions are exactly F and $\hat{F}$.

Economic message: Figure 4 turns the theoretical conjugacy test into a construction procedure for optimal private private structures.

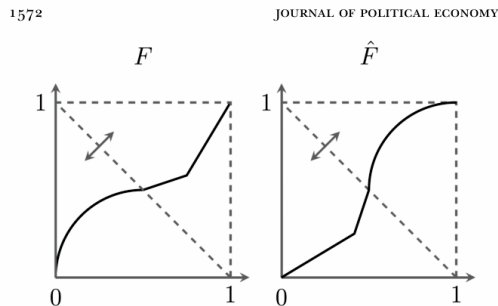


FIG. 3.—Example of cumulative distribution function F and its conjugate $\hat{F}$. Shapes under the curves are congruent: the transformation that maps one to the other is reflection around the antidiagonal. Qualitatively, F corresponds to the belief distribution of a more informative signal, and $\hat{F}$ corresponds to that of a less informative signal (because the former assigns less mass to posterior beliefs near 0.5).

(a) Figure 3: CDF and conjugate CDF

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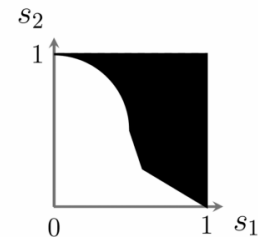


FIG. 4.—Private private information structure, where beliefs $p(s_1)$ and $p(s_2)$ are distributed according to the pair of conjugate distributions F and $\hat{F}$ from figure 3: the signals are uniform on $[0, 1]^2$, and $\omega = 1$ if and only if $s_2 \geq \hat{F}(1 - s_1)$ (black region).

(b) Figure 4: Structure from conjugate distributions

5.5 Figure 5: Conjugate of a discrete distribution

Read:

- Figure 5 shows how conjugacy works for a discrete belief distribution.
- Atoms in F become zero-mass intervals in $\hat{F}$.
- Zero-mass intervals in F become atoms in $\hat{F}$.
- A distribution with three interior atoms can have a conjugate with four atoms because of endpoint behavior.

Economic message: The figure makes clear why binary-valued or finitely valued signals can still generate nontrivial privacy-preserving recommendations.

5.6 Figure 6: Sets of uniqueness and non-uniqueness

Read:

- Panel A displays a set of uniqueness, with an upward-closed black region.
- Panels B and C show different black regions with the same projections onto the axes.
- Because B and C have identical projections but different sets, they are not sets of uniqueness.
- The figure illustrates why projections determine belief distributions but may not always determine the full information structure.

Economic message: Figure 6 explains the connection to tomography: if projections pin down the set, the information structure is frontier-optimal.

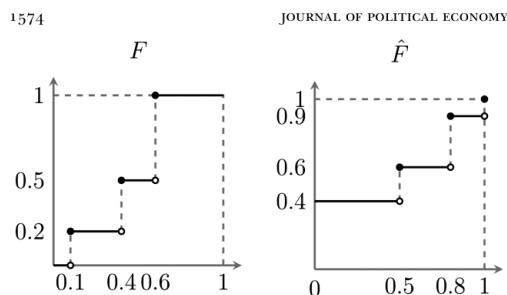


FIG. 5.—Conjugate of discrete distribution F with three atoms, at 0.1, 0.4, and 0.6. Each atom becomes an interval of zero measure with the length equal to the atom's weight, and vice versa. Since F does not have atoms at the endpoints of $[0, 1]$, the number of intervals of zero measure exceeds the number of atoms by 1, so its conjugate $\hat{F}$ has four atoms, at 0, 0.5, 0.8, and 1.

(a) Figure 5: Discrete conjugate distribution

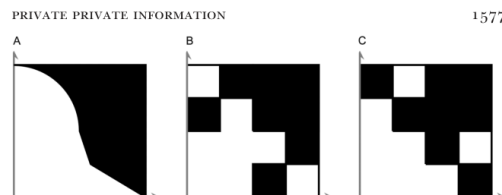


FIG. 6.—A, Black area is a set of uniqueness, since it is upward closed (theorem 3). B, C, Black areas are not sets of uniqueness, since they are different but have the same projections on the axes.

(b) Figure 6: Sets of uniqueness

5.7 Figure 7: Dominant privacy-preserving recommendation

Read:

- Figure 7 studies the case where the protected signal s_1 is a symmetric binary signal matching the state with probability $3/4$.
- Panel A shows the CDF F of posteriors induced by s_1 .
- Panel B shows the conjugate distribution for the dominant recommendation.
- The dominant recommendation is trinary: it fully reveals the state with probability $1/2$ and is completely uninformative with probability $1/2$.

Economic message: This is the main applied privacy lesson. The optimal privacy-preserving recommendation sometimes randomizes between full disclosure and no disclosure.

5.8 Figure 8: Social welfare-maximizing private information structure

Read:

- Figure 8 gives the welfare-maximizing *private private* structure for a two-agent decision problem with quadratic-type payoff $1 - 2|\omega - a|$.
- The black region is chosen to maximize the designer's welfare subject to the independence/privacy constraint.
- The figure shows that the optimal design need not coincide with the simplest diagonal or binary examples.
- It illustrates how the geometry of the set changes with the designer's objective.

Economic message: Once a designer has a particular welfare function, the frontier can be optimized further. Privacy narrows the feasible set, but the designer still chooses among frontier structures.

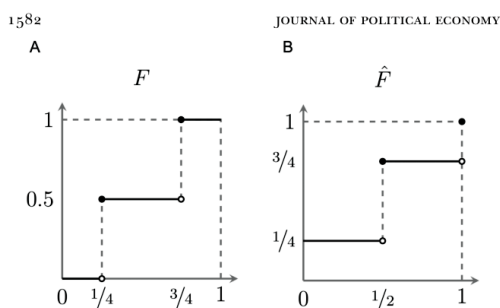


FIG. 7.—Dominant privacy-preserving recommendation when $3/4$ -binary signal s_1 must be kept secret. A, Cumulative distribution function F of posteriors induced by the symmetric binary signal s_1 matching the state with probability $3/4$. B, Dominant recommendation s_2^* corresponds to the conjugate distribution $\hat{F}$. We see that s_2^* is trinary: it is completely uninformative with probability $1/2$ and fully reveals the state with the complementary chance, inducing the posteriors 0 or 1 with equal probabilities.

(a) Figure 7: Dominant privacy-preserving recommendation

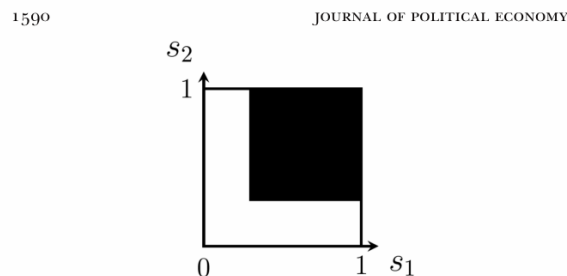


FIG. 8.—Social welfare-maximizing private private information structure for the decision problem in which $u_1(\omega, a) = u_2(\omega, a) = 1 - 2|\omega - a|$.

assume $\beta \geq \alpha$ without loss of generality, and hence the middle atom of $\hat{\mu}$ is above $1/2$. Therefore, the second agent contributes $1 - 2\alpha$ to the wel-

(b) Figure 8: Welfare-maximizing private private structure

5.9 Table 1: Family of information structures I_ε indexed by ε

Read:

- Table 1 gives a family of binary-state, binary-signal information structures indexed by $\varepsilon \in [0, 1/4]$.
- At $\varepsilon = 0$, the structure preserves informational rents in the buyer example because the seller cannot use one buyer's signal to infer the other's.
- For any $\varepsilon > 0$, small correlation allows a seller to cross-infer and extract full surplus in the common-value environment discussed around the table.
- The table demonstrates how fragile privacy can be: an arbitrarily small departure from independence can have large strategic consequences.

Economic message: The table motivates why exact *private private* constraints matter in mechanism design. Correlation across receivers can radically change surplus extraction.

TABLE 1
FAMILY OF INFORMATION STRUCTURES I_ε
INDEXED BY $\varepsilon \in [0, 1/4]$

	$s_2 = 1$	$s_2 = 0$
$\omega = 1$:		
$s_1 = 1$	$.5 + 2\varepsilon$	$.25 - \varepsilon$
$s_1 = 0$	$.25 - \varepsilon$	0
$\omega = 0$:		
$s_1 = 1$	0	$.25 - \varepsilon$
$s_1 = 0$	$.25 - \varepsilon$	$.5 + 2\varepsilon$

Table 1: Family of information structures I_ε

6 Short summary

- The paper defines *private private* information structures: signals are independent but informative about a common state.
- It measures informativeness using Blackwell dominance and defines Blackwell–Pareto optimality as a frontier under the privacy constraint.
- For binary state and two signals, Blackwell–Pareto optimality is equivalent to conjugacy of posterior-belief distributions.
- This implies a sharp trade-off: making one independent signal more informative forces the other to become less informative.
- Every *private private* structure is equivalent to one generated by a partition of a unit cube.
- For binary states and many signals, Blackwell–Pareto optimality is equivalent to the associated set being a set of uniqueness.
- In privacy-preserving recommendation with a binary state, there is a dominant recommendation independent of the protected signal.
- The dominant recommendation’s posterior-belief distribution is the conjugate of the protected signal’s posterior-belief distribution.
- The paper provides a simple sampling procedure for constructing the dominant recommendation.
- Privacy imposes information loss: independent signals cannot duplicate the same bits of state information.
- The framework connects statistical parity in fairness to Blackwell information design.
- The same mathematics applies to information design across receivers, causal colliders, mechanism-design feasibility, and persuasion in zero-sum games.

7 Conclusion

7.1 Core conclusion

He, Sandomirskiy, and Tamuz develop a theory of information that is simultaneously private across signals and informative about a common state. The paper’s central insight is that independence across signals creates a genuine information frontier. One cannot freely provide all agents with highly informative signals while keeping the signals independent of one another.

7.2 Economic intuition

The economic intuition is that signals about a common state naturally become correlated through the state. If a recommendation is strongly informative about a state that is correlated with a sensitive attribute, the recommendation will generally leak information about that attribute. The only way to preserve privacy is to distort, randomize, or coarsen the recommendation. The question is how to do this without wasting more information than necessary.

The paper’s answer is to use the Blackwell order. Rather than choosing a specific loss function or application, the authors ask for signals that are not dominated in any decision problem. This makes the frontier robust and allows the theory to apply across recommendation, fairness, information design, and mechanism design.

7.3 Why the conjugacy result matters

The conjugacy theorem is the paper’s central practical tool. In the binary-state, two-signal case, it gives a complete and visual recipe for optimality: the two posterior-belief distributions must be conjugates. If the protected signal induces very informative beliefs, the recommendation must induce a less informative conjugate belief distribution. If the protected signal is weak, the recommendation can be stronger.

This is useful for applications such as credit scoring, hiring recommendations, or algorithmic classification under statistical parity. A designer can first estimate the posterior beliefs induced by protected attributes and then construct the most informative independent recommendation by taking the conjugate distribution.

7.4 Implications for fairness and privacy

The fairness interpretation is especially important. Statistical parity requires recommendations to be independent of protected attributes. The paper shows that this constraint is not merely a constraint on group averages. It is an information constraint that limits how much state-relevant information can be communicated. The framework therefore provides a way to quantify the efficiency cost of statistical parity and to design recommendations that attain the best possible Blackwell informativeness under that constraint.

7.5 Implications for multiagent information design

The results also explain why privacy across receivers is strategically important. In common-value environments, even tiny correlation across receivers’ signals can allow a seller to extract surplus using one receiver’s information against another. Table 1 illustrates that strategic consequences can change discontinuously when privacy is violated. The *private private* constraint is therefore not only about fairness or confidentiality; it also changes the feasible surplus division in games and markets.

7.6 Broader methodological contribution

A major contribution is methodological. The paper connects economic information design to mathematical tomography. Belief distributions are projections of geometric sets, and optimality corresponds to uniqueness from projections. This is an unusual but powerful connection: it converts questions about optimal signals into questions about whether geometric objects are identified by their shadows.

7.7 Limitations and open questions

The strongest constructive results are for binary states. With three or more states, optimal privacy-preserving recommendations can depend on the decision-maker’s objective, so no universal dominant recommendation need exist. The exact independence requirement may also be too strict in some empirical or algorithmic

applications, where approximate privacy is acceptable. Finally, the paper is theoretical; implementing its recommendations requires estimating the joint distribution of the state and protected variables.

7.8 Takeaway

The main takeaway is that privacy has a precise information-theoretic price. The paper identifies the frontier of that price in important cases and gives constructive methods for attaining it. It shows that the right privacy-preserving signal is not simply the most garbled signal or the least discriminating signal. It is the signal whose posterior distribution is optimally paired with the protected information through conjugacy.